

5 Joint Probability Distributions

5.1 Joint Probability Distributions for Two Random Variables

Definition. Let X, Y be two random variables. With a ordered pair (X, Y) , we define a **joint probability distribution function** and denote

$$f_{XY}(x, y).$$

In discrete case, we called **joint probability mass function**; In continuous case, we called **joint probability density function**.

Property. Let X, Y be two random variables.

1. If $f_{XY}(x, y)$ is a joint probability mass function, then it satisfies the following properties:

$$(1) f_{XY}(x, y) \geq 0$$

$$(2) \sum_{x \in X} \sum_{y \in Y} f_{XY}(x, y) = 1$$

$$(3) f_{XY}(x, y) = P(X = x, Y = y) = P(X = x \text{ \& } Y = y)$$

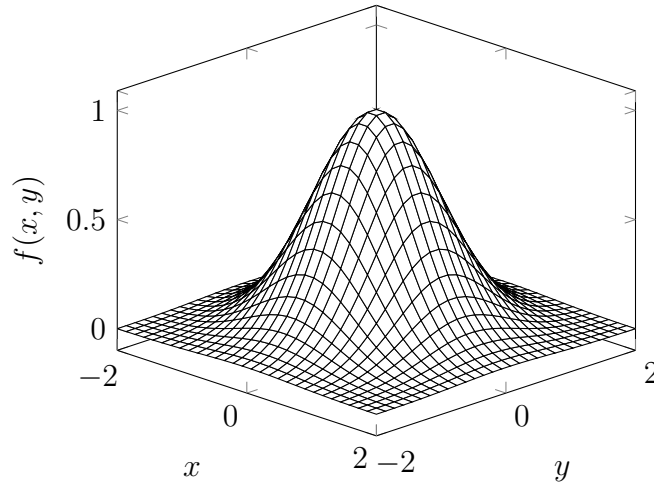
2. If $f_{XY}(x, y)$ is a joint probability density function, then it satisfies the following properties:

$$(1) f_{XY}(x, y) \geq 0 \text{ for all } x, y$$

$$(2) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$$

$$(3) \text{ For any region } R \text{ of two-dimensional space,}$$

$$f_{XY}(x, y) = P((X, Y) \in \mathbb{R}) = \iint_R f_{XY}(x, y) dx dy.$$



Definition. (Marginal Density) Let X, Y be two random variables.

1. For discrete case, the marginal(邊際) density is defined as

$$(1) f_X(x) = \sum_{y \in Y} f_{XY}(x, y)$$

$$(2) f_Y(y) = \sum_{x \in X} f_{XY}(x, y)$$

2. For continuous case, the marginal density is defined as

$$(1) f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$(2) f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

Example. Suppose $f_{XY}(x, y)$ is a joint probability mass function with two random variables $X = \{0, 1, 2\}$ and $Y = \{0, 1, 2, 3\}$. It is as shown in the table below

$Y = y$	$X = x$		
	0	1	2
0	0.15	0.1	0.05
1	0.02	0.1	0.05
2	0.02	0.03	0.2
3	0.01	0.02	0.25

Then the marginal probability distributions of X and Y are

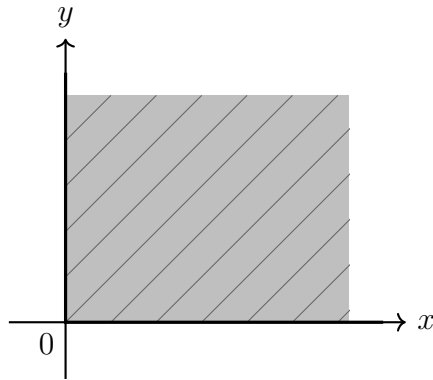
$Y = y$	$X = x$			
	0	1	2	$f_Y(y)$
0	0.15	0.1	0.05	0.3
1	0.02	0.1	0.05	0.17
2	0.02	0.03	0.2	0.25
3	0.01	0.02	0.25	0.28
$f_X(x)$	0.2	0.25	0.55	1

Example. Let $f_{XY}(x, y)$ be a joint probability density function and be defined as

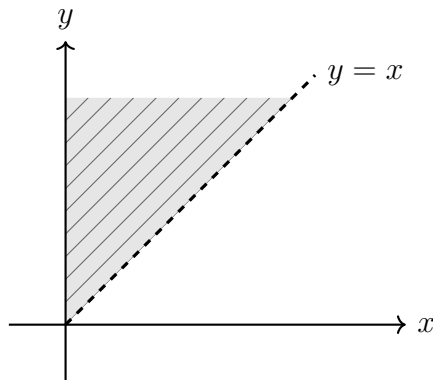
$$f_{XY}(x, y) = ke^{-0.01x-0.02y} \text{ for } 0 < x < y < \infty,$$

where $k = 6 \times 10^{-6}$. The region with nonzero probability is shaded in below. Find the $P(Y > 2000)$.

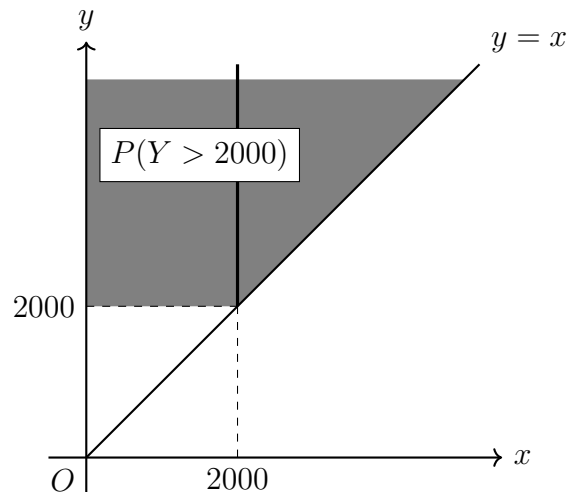
(i) **Region :** $0 < x < \infty, 0 < y < \infty$



(ii) **Region :** $0 < x < y < \infty$



Sol. The region is shaded in below.



So

$$\begin{aligned}
P(Y > 2000) &= \int_{2000}^{\infty} \int_0^y f_{XY}(x, y) dx dy \\
&= \int_{2000}^{\infty} \int_0^y k e^{-0.01x - 0.02y} dx dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \left[\int_0^y e^{-0.01x} dx \right] dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \left[\frac{e^{-0.01x}}{-0.01} \right]_0^y dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \cdot 100(1 - e^{-0.01y}) dy \\
&= 100k \int_{2000}^{\infty} (e^{-0.02y} - e^{-0.03y}) dy \\
&= 100k \lim_{b \rightarrow \infty} \left[\frac{e^{-0.02y}}{-0.02} - \frac{e^{-0.03y}}{-0.03} \right]_{2000}^b \\
&= 100k \left[(0 - 0) - \left(\frac{e^{-0.02(2000)}}{-0.02} - \frac{e^{-0.03(2000)}}{-0.03} \right) \right] \\
&= 100k \left(50e^{-40} - \frac{100}{3}e^{-60} \right)
\end{aligned}$$

Substitute $k = 6 \times 10^{-6}$:

$$\begin{aligned}
P(Y > 2000) &= 6 \times 10^{-4} \left(50e^{-40} - \frac{100}{3}e^{-60} \right) \\
&= 0.03e^{-40} - 0.02e^{-60}.
\end{aligned}$$

5.2 Conditional Probability Distributions and Independence

5.2.1 Independence

Definition. Let X and Y be two random variables. Then they are **independent** if and only if

$$f_{XY}(x, y) = f_X(x)f_Y(y).$$

Remark. Recall math in high school, we know that

$$P(A \cap B) = P(A)P(B)$$

if both events A and B are independent.

Definition. Let X and Y be two random variables. Then the mean from a joint distribution (X, Y) with a transformer function $H(X, Y)$ of X and Y is defined as

$$E[H(X, Y)] = \sum_{x \in X} \sum_{y \in Y} H(X, Y) f_{XY}(x, y) \quad (\text{Discrete case})$$

$$E[H(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} H(X, Y) f_{XY}(x, y) dx dy \quad (\text{Continuous case})$$

Example. Find $\sum_{x \in X} \sum_{y \in Y} x f_{XY}(x, y)$.

Sol.

$$\begin{aligned} \sum_{x \in X} \sum_{y \in Y} x f_{XY}(x, y) &= \sum_{x \in X} x \left(\sum_{y \in Y} f_{XY}(x, y) \right) \\ &= \sum_{x \in X} x f_X(x) \\ &= E(X). \end{aligned}$$

Example. Find $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x, y) dy dx$.

Sol.

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x, y) dy dx &= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{XY}(x, y) dy \right) dx \\ &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= E(X). \end{aligned}$$

5.2.2 Covariance and Correlation

Let's review the variance in one variable.

$$\begin{aligned} \text{Var}(X) &= E((X - E(X))^2) \\ &= E(X^2 - 2XE(X) + E^2(X)) \\ &= E(X^2) - 2E^2(X) + E^2(X) \\ &= E(X^2) - E^2(X) \end{aligned}$$

So we have

$$\begin{aligned} \text{Var}(X) &= E(X^2) - E^2(X) \\ &= E(X \cdot X) - E(X) \cdot E(X) \\ &\Rightarrow E(X \cdot Y) - E(X) \cdot E(Y). \end{aligned}$$

$E(X \cdot Y) - E(X) \cdot E(Y) = E[(X - E(X))(Y - E(Y))]$, it is covariance(共變異數).

Definition. The **covariance** between the random variables X and Y , denoted as $\text{cov}(X, Y)$ or σ_{XY} is

$$\text{cov}(X, Y) \equiv E[(X - E(X))(Y - E(Y))] = E(XY) - E(X)E(Y).$$

Theorem. Let (X, Y) be a ordered pair random variables with joint probability distribution function $f_{XY}(x, y)$. If X and Y are independent, then $\text{cov}(X, Y) = 0$.

Definition. (Correlation coefficient) The **Pearson correlation coefficient**(皮爾森相關係數) between random variables X and Y , denoted as ρ_{XY} , is

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

For any two random variables X and Y ,

$$-1 \leq \rho_{XY} \leq 1.$$

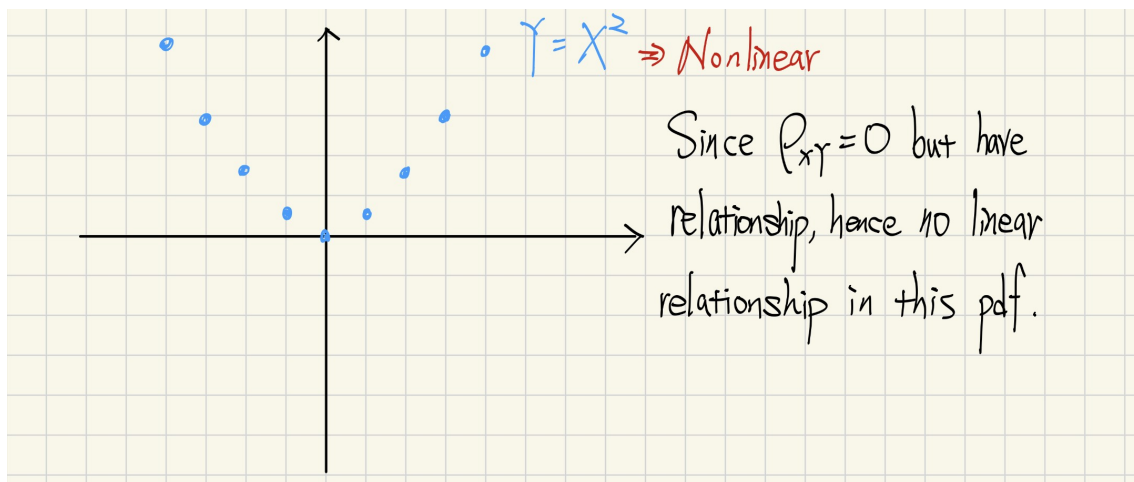
Theorem. The conditions for **linearity** between X and Y if and only if

$$Y = \beta_0 + \beta_1 X.$$

Remark. Let ρ_{XY} be the Pearson correlation coefficient between random variables X and Y .

1. If $\rho_{XY} = 1$, then we call it is perfect positive correlation and $\beta_1 > 0$ for $Y = \beta_0 + \beta_1 X$.
2. If $\rho_{XY} = -1$, then we call it is perfect negative correlation and $\beta_1 < 0$ for $Y = \beta_0 + \beta_1 X$.
3. If $\rho_{XY} = 0$, then we call it is no linear relationship.

Example. Consider $Y = X^2$.



Remark. If X and Y are independent, then $\rho_{XY} = 0$.

5.2.3 Conditional Probability Distributions

Definition. Let (X, Y) be a ordered pair random variables with joint probability distribution function $f_{XY}(x, y)$. Suppose $f_X(x)$ and $f_Y(y)$ are marginal probability distribution function for $f_{XY}(x, y)$.

- (1) The conditional density for X given $Y = y$ is defined as

$$f_{X|y}(x) = \frac{f_{XY}(x, y)}{f_Y(y)}.$$

Recall math in high school, we know that

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

- (2) The conditional density for Y given $X = x$ is defined as

$$f_{Y|x}(y) = \frac{f_{XY}(x, y)}{f_X(x)}.$$

Example. Suppose

$$f_{XY}(x, y) = \frac{c}{x}, \quad c = \frac{1}{6 - 27 \ln(33/27)},$$

where $27 \leq y \leq x \leq 33$. Find

- (a) $f_X(x)$
- (b) $f_Y(y)$
- (c) $f_{X|y}(x)$
- (d) The probability that $x > 32$ given $y = 30$.
- (e) The expected value of X given $y = 30$.

Sol. We don't substitute c into the function since it is too difficult to evaluate.

(a)

$$\begin{aligned} f_X(x) &= \int_{27}^x f_{XY}(x, y) dy \\ &= \int_{27}^x \frac{c}{x} dy \\ &= \left. \frac{cy}{x} \right|_{27}^x \\ &= c \frac{x - 27}{x}, \text{ where } 27 \leq x \leq 33. \end{aligned}$$

(b)

$$\begin{aligned} f_Y(y) &= \int_y^{33} f_{XY}(x, y) dx \\ &= \int_y^{33} \frac{c}{x} dx \\ &= c(\ln 33 - \ln y) \\ &= c \ln \frac{33}{y}, \text{ where } 27 \leq y \leq 33. \end{aligned}$$

(c)

$$\begin{aligned} f_{X|y}(x) &= \frac{f_{XY}(x, y)}{f_Y(y)} \\ &= \frac{c/x}{c(\ln 33 - \ln y)} \\ &= \frac{1}{x(\ln 33 - \ln y)} \\ &= \frac{1}{x \ln(33/y)}, \text{ where } y \leq x \leq 33. \end{aligned}$$

(d)

$$\begin{aligned} P(X > 32|y = 30) &= \int_{32}^{33} \frac{1}{x(\ln 33/30)} dx \\ &= \left[\frac{\ln |x|}{\ln(33/30)} \right]_{32}^{33} \\ &= \frac{\ln 33 - \ln 32}{\ln(33/30)} \\ &= \frac{\ln(33/32)}{\ln(33/30)}. \end{aligned}$$

(e)

$$\begin{aligned} E(X|y = 30) &= \int_{30}^{33} x f_{X|y=30}(x) dx \\ &= \int_{30}^{33} x \frac{1}{x \ln(33/30)} dx \\ &= \int_{30}^{33} \frac{1}{\ln(33/30)} dx \\ &= \frac{3}{\ln(33/30)}. \end{aligned}$$

5.3 Joint Probability Distributions for More Than Two Random Variables

Definition. A **joint probability density function** for the continuous random variables $X_1, X_2, \dots, X_p$, denoted as $f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p)$, satisfies the following properties:

- (1) $f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) \geq 0$
- (2) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) dx_1 dx_2 \dots dx_p = 1$
- (3) For any region B of p -dimensional space,

$$P((x_1, x_2, \dots, x_p) \in B) = \int \dots \int_B f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) dx_1 \dots dx_p$$

Definition. The marginal density of X_1 for joint probability density function

$$f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p)$$

is defined as

$$f_{X_1}(x_1) = \int \dots \int_{\bigcup_{k=2}^p X_k} f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) dx_2 dx_3 \dots dx_p.$$

Definition. From above definition, we can define mean and variance that is

$$\begin{aligned} E(X_1) &= \int \dots \int_{\bigcup_{k=2}^p X_k} x_1 f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) dx_2 dx_3 \dots dx_p \\ &= \int_{-\infty}^{\infty} x_1 f_{X_1}(x_1) dx_1 \end{aligned}$$

and

$$Var(X_1) = \int_{-\infty}^{\infty} [x_1 - E(X_1)]^2 f_{X_1}(x_1) dx_1.$$

Remark. Conditional probability distributions can be developed for multiple random variables by an extension of the ideas used for two random variables. for example, the joint conditional probability distribution of X_1, X_2 and X_3 given $X_4 = x_4$ and $X_5 = x_5$ is

$$f_{X_1 X_2 X_3 | x_4 x_5}(x_1, x_2, x_3) = \frac{f_{X_1 X_2 X_3 X_4 X_5}(x_1, x_2, x_3, x_4, x_5)}{f_{X_4 X_5}(x_4, x_5)}$$

for $f_{X_4 X_5}(x_4, x_5) > 0$.

Example. Suppose that the random variables X , Y and Z have joint probability density function

$$f_{XYZ}(x, y, z) = c$$

over the cylinder $x^2 + y^2 < 4$ and $0 < z < 4$.

- (1) Determine the constant c so that $f_{XYZ}(x, y, z)$ is a probability distribution function.
- (2) Find $P(X^2 + Y^2 < 2)$
- (3) Find $P(Z < 2)$
- (4) Find $E(X)$
- (5) Find $P(X < 1|Y = 1)$
- (6) Find $P(X^2 + Y^2 < 1|Y = 1)$
- (7) Find the conditional probability distribution of Z given that $X = 1$ and $Y = 1$.

Sol.

- (1) We want to find c such that $\iiint_T c \, dx \, dy \, dz = 1$.

Since we know that it is over the cylinder $x^2 + y^2 < 4$ and $0 < z < 4$.

Let $T = \{(x, y, z) \mid x^2 + y^2 < 4, 0 < z < 4\}$. This implies

$$\begin{aligned} \iiint_T c \, dx \, dy \, dz &= c \iiint_T 1 \, dx \, dy \, dz \\ &= c \iiint_T 1 \, dV \\ &= c \times 2^2 \times 4 \times \pi \\ &= 16\pi \times c = 1 \end{aligned}$$

$$\text{Hence } c = \frac{1}{16\pi}.$$

- (2) Let $T = \{(x, y, z) \mid 0 < z < 4, -\sqrt{2} < y < \sqrt{2}, -\sqrt{2-y^2} < x < \sqrt{2-y^2}\}$.

$$\begin{aligned} P(X^2 + Y^2 < 2) &= \iiint_T c \, dx \, dy \, dz \\ &= c \int_0^4 \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-y^2}}^{\sqrt{2-y^2}} 1 \, dx \, dy \, dz \\ &= c \int_0^4 dz \int_{-\sqrt{2}}^{\sqrt{2}} 2\sqrt{2-y^2} \, dy \\ &= 8c \int_{-\sqrt{2}}^{\sqrt{2}} \sqrt{2-y^2} \, dy \\ &= 8 \times \frac{1}{16\pi} \times \pi = \frac{1}{2}. \end{aligned}$$

(3)

$$P(Z < 2) = c \iiint_T dV \times \frac{1}{2} = \frac{1}{2}.$$

(4) Let $T = \{(x, y, z) \mid 0 < z < 4, -\sqrt{4-x^2} < y < \sqrt{4-x^2}, -2 < x < 2\}$.

$$\begin{aligned} E(X) &= \int_{-2}^2 x f_X(x) dx \\ &= \int_{-2}^2 x \left[\int_0^4 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} c dy dz \right] dx \\ &= 4c \int_{-2}^2 x \cdot 2\sqrt{4-x^2} dx \\ &= 8c \int_{-2}^2 x\sqrt{4-x^2} dx \\ &= 8c \left[\frac{-1}{3} u^{3/2} \right]_0^0 = 0. \end{aligned}$$

(5) Let $R = \{(x, z) \mid 0 < z < 4, -\sqrt{4-y^2} < x < \sqrt{4-y^2}\}$.
Compute

$$\begin{aligned} f_{XZ|Y=1}(x, z) &= \frac{c}{f_Y(y=1)} \\ &= c \left/ \iint_R c dx dz \right|_{y=1} \\ &= \left[\int_0^4 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 1 dx dz \right]_{y=1}^{-1} \\ &= \left[\int_0^4 \int_{-\sqrt{3}}^{\sqrt{3}} 1 dx dz \right]^{-1} \\ &= \frac{1}{8\sqrt{3}} \end{aligned}$$

Since it is given $Y = 1$, we have

$$x^2 + 1 < 4 \iff -\sqrt{3} < x < \sqrt{3}.$$

Suppose $\Omega = \{(x, z) \mid -\sqrt{3} < x < 1, 0 < z < 4\}$.

And thus

$$\begin{aligned} P(X < 1 | Y = 1) &= \iint_{\Omega} f_{XZ|Y=1}(x, z) dx dz \\ &= \int_0^4 \int_{-\sqrt{3}}^1 \frac{1}{8\sqrt{3}} dx dz \\ &= 4 \times \frac{1 + \sqrt{3}}{8\sqrt{3}} = \frac{1 + \sqrt{3}}{2\sqrt{3}}. \end{aligned}$$

- (6) Given $x^2 + y^2 < 1$ and $Y = 1$, we obtain $x^2 + 1 < 1 \iff x^2 < 0$. Since there is no real number x satisfying this inequality, the probability is 0. Hence

$$P(X^2 + Y^2 < 1 | Y = 1) = 0.$$

- (7) Compute

$$P_{Z|x=1, y=1}(z) = \frac{c}{f_{XY}(x=1, y=1)} = \frac{1}{4}.$$

5.4 Common Joint Distributions

5.4.1 Multinomial Probability Distribution

Remark. Recall binomial distribution, let X be the number of trials that result in a success $x = 0, 1, \dots, n$, $n = 1, 2, 3, \dots$ with the parameter $p \in (0, 1)$. Then

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

Definition. Suppose that a random experiment consists of a series of n trials. Assume that

- (1) The result of each trial is classified into one of k classes.
- (2) The probability of a trial generating a result in class 1, class 2, $\dots$, class k is $p_1, p_2, \dots, p_k$, respectively, and these probabilities are constant over the trials.
- (3) The trials are independent.

The random variables $X_1, X_2, \dots, X_k$ hat denote the number of trials that result in class 1, class 2, $\dots$, class k , respectively, have a **multinomial distribution**(多項分布) and the joint probability mass function is

$$P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

for $x_1 + x_2 + \dots + x_k = n$ and $p_1 + p_2 + \dots + p_k = 1$.

Theorem. If $X_1, X_2, \dots, X_k$ have a multinomial distribution, the marginal probability distribution of X_i is binomial with

$$E(X_i) = np_i$$

and

$$V(X_i) = np_i(1 - p_i).$$

5.4.2 Bivariate Normal Probability Density Function

Remark. Recall normal distribution, we know

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where $-\infty < x < \infty$.

Definition. The probability density function of a **bivariate normal distribution**(二變數常態分佈) is

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^u$$

for

$$u = \frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right],$$

where

$$\begin{aligned} -\infty < x < \infty, & & -\infty < y < \infty, \\ \sigma_X > 0, & & -\infty < \mu_X < \infty, \\ \sigma_Y > 0, & & -\infty < \mu_Y < \infty, \\ -1 < \rho < 1. \end{aligned}$$

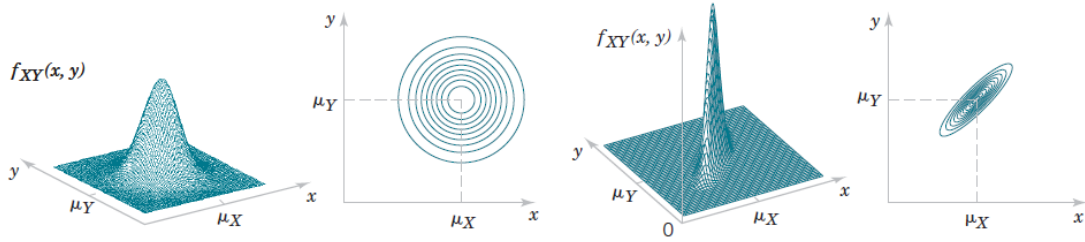


FIGURE 5.12

Examples of bivariate normal distributions.

Theorem. Suppose X and Y are bivariate normal distribution. If $\rho = 0$, then X and Y are independent.

In other view, we can use a matrix to define bivariate normal distribution and even more generalizing.

Definition. The probability density function of a multivariate normal distribution is

$$f_X(x_1, \dots, x_k) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu) \right)}$$

for Σ is a covariance matrix of $X_1, \dots, X_k$.

Remark. In bivariate case, it is defined

$$\mu = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}.$$